

재료공학실험(공) (MSE307-01)

학기	2026년 1학기
담당교수	김진수
이수구분	교선평수
학점	3.0
강의시간	수16:30-19:30(E6 - 223), 금13:00-16:00(E6 - 223)
강의실	
소속	기초학부

교과개요

This is a hands-on course that acquires essential experimental techniques in the material research and development process through experiments on the synthesis, processing, and property evaluation of various materials (metals, ceramics, polymers, semiconductors, nanomaterials, etc.). In this course, you will analyze the physical, mechanical, thermal, electrical, and optical properties of materials and learn measurement techniques for research and industrial applications. The experimental course is structured to reflect basic materials engineering knowledge and the latest research trends, and each experiment is conducted in units of 2 to 3 weeks. Through this, students will cultivate the ability to design experiments, analyze data, write reports, and learn experimental approaches that can be utilized in actual research and industrial environments.

교과목표

1. Understanding material synthesis and processing processes of ceramics, polymers, metals/semiconductors, and nanomaterials, and understand the effects of each process on the properties of the material. 2. Acquire material property evaluation and measurement techniques such as structural, mechanical, thermal, electrical, and optical properties. 3. Understand the property analysis and application of various materials through each experiment, analyze the relationship between microstructure and properties. 4. Improve experimental data analysis and research report writing skills with quantitative analysis, error interpretation, and statistical processing methods of experimental data, and the ability to write systematic research reports. 5. Cultivating problem-solving and teamwork skills through the experience of solving problems that arise during the experiment.

주차별 강의계획

1주차:Introduction
2주차:Experimental project 1-1
3주차:Experimental project 1-2
4주차:Experimental project 1-3
5주차:Experimental project 2-1
6주차:Experimental project 2-2
7주차:Experimental project 2-3
8주차:Midterm project report
9주차:Experimental project 3-1
10주차:Experimental project 3-2
11주차:Experimental project 3-3
12주차:Experimental project 4-1
13주차:Experimental project 4-2
14주차:Experimental project 4-3

15주차:Group presentation

16주차:Final project report

학습윤리

Unauthoritarian assistance is not allowed on all assignments and exams. Misrepresentation of your identity is not also permitted.

교과정책(방침)

The course will consist mainly of experimental projects. Students are required to submit assignments on the given date, and late submissions will result in a deduction of points. Additional materials (papers, experimental results, etc.) will be provided when necessary.